Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–July 11th, 2023

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. How the time complexity in a PRAM model is computed? (1)

B. How is the communication via shared memory performed in a PRAM model? (latency vs parallel accesses) (1)

C. Discuss the following sentence: "If a problem has no feasible/efficient solution on PRAM, it has no feasible/efficient solution for any parallel machine." True or False? (1)

D. Define the "work" in a PRAM model. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss how the complexity of the powering of a number algorithm can be computed. Please report also the complexity in terms of time and space. (2)

B. The best worst-case running time that we have seen for comparison sorting is $O(n \lg n)$. Can you explain why? (2)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain how the disjoint set data structure works and which optimizations have been considered to reduce the complexity of the operations. (2) (You can skip this question if you completed the first challenge)

B. Describe the skip list data structure and how the union is performed, and which complexity has. (2)

Exercise n. 4

Answer the following questions about the amortized analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Detail the Longest Common Subsequence algorithm: pseudo code and space and temporal complexity. (2)

B. What is meant by "accounting method"? At which class of algorithms could be applied? (2)